

MUHAMMED SAAHIL

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Summary

Enthusiastic and curious problem solver with a strong passion for AI and software development, actively building real-world end-to-end systems that solve meaningful problems. A continuous learner eager to grow, contribute, and make an impact in a software engineering role.

Skills

Languages: Python, SQL, HTML

Machine Learning: TensorFlow, Keras, Scikit-learn, PyTorch, NumPy, Pandas, MLflow, DagsHub, Supervised Learning, Unsupervised Learning, Feature Engineering, Model Training & Evaluation, Jupyter Notebook

Deep Learning: CNN, RNN, LSTM, Neural Networks, Natural Language Processing

Backend Development: Flask, FastAPI, Streamlit, REST API Development

Databases & Storage: MongoDB, SQLite3, MySQL, AWS S3

Cloud & DevOps: AWS (EC2, S3, Elastic Beanstalk), Docker, Git/GitHub, CI/CD

Other Skills: N8n Automations, Adobe Photoshop, Adobe Illustrator, Microsoft Office Suite, Technical Documentation

Project Work

- **ML-Powered Network Intrusion Detection System | End-to-End MLOps:**
 - Developed end-to-end ML system for network security with automated pipeline: data preprocessing → model training → deployment → monitoring
 - Built ELT pipeline and trained classification model achieving 95%+ accuracy on intrusion detection
 - Deployed FastAPI-based inference service on AWS EC2 with real-time prediction capabilities
 - Implemented CI/CD automation using GitHub Actions and Docker for reproducible deployments
 - Integrated MLflow for experiment tracking, DAGsHub for pipeline orchestration, and MongoDB for Database
- **Skin Cancer Detection System | Deep Learning:**
 - Built a binary classification CNN model to detect Benign vs Malignant skin cancer from medical images achieving 84-85%+ accuracy
 - Applied data augmentation and normalization using ImageDataGenerator to improve model generalization and prevent overfitting
 - Designed CNN architecture using Keras Sequential API with Conv2D, MaxPooling, Dense, and Dropout layers
 - Deployed real-time prediction web application using Streamlit with image upload and instant classification output
- **AI Powered RAG Chatbot for Company Policies | GenAI, LangChain:**
 - Built a RAG-based chatbot using LangChain and Gemini LLM that answers employee queries strictly from company policy documents preventing hallucinated responses
 - Used Sentence Transformers to convert policy documents into vector embeddings stored in a FAISS index enabling accurate semantic search across company policies
 - Integrated LangChain retriever with Gemini API to fetch relevant policy chunks and generate grounded, context-aware responses for each user query

Experience

Voila System Pvt.Ltd

Mar 2024 - Jul 2024

Software Intern

- Built and optimized applications using Python and MySQL, ensuring efficient query
- Applied data structures and algorithms in problem-solving tasks to improve performance
- Mentored fellow interns on debugging, teamwork, and effective use of AI tools

Lanovis Security Solutions

Jul 2025 - Dec 2025

AI Intern

- Developed machine learning and deep learning models using TensorFlow and PyTorch, gaining experience with neural networks (CNN, RNN, LSTM), NLP
- Built AI chatbot using LangChain for internal automation and assisted in development of company AI products
- Learned professional development practices including team collaboration, version control, agile methodologies, and production deployment workflows

Education

Bachelor of Technology in Computer Science and Engineering

Aug 2021 - Jun 2025

SCMS School of Engineering and Technology